

KORIJEŃOVANJE KOMPLEKSNIH BROJEVA:

$$z = r(\cos(\varphi) + i\sin(\varphi))$$

$$\left(\sqrt[m]{z}\right)_k = \sqrt[m]{r} \left(\cos\left(\frac{\varphi + 2k\pi}{m}\right) + i\sin\left(\frac{\varphi + 2k\pi}{m}\right) \right)$$

k - "rotacijski broj" $\hat{=} k = 0, 1, 2, 3, \dots, m-1$.

Pr. $m=7$ $\hat{=} k = 0, 1, 2, 3, 4, 5, 6$

EXPONENCIJALNI OBLIK:

$$z = r(\underbrace{\cos(\varphi) + i\sin(\varphi)}_{e^{i \cdot \varphi}})$$



$$z = r \cdot e^{i \cdot \varphi}$$

"Eulerov zapis"

RAČUNSKIE OPERACIJE EXP. OBLIKA:

$$z_1 = r_1 \cdot e^{i \cdot \varphi_1}$$

$$z_2 = r_2 \cdot e^{i \cdot \varphi_2}$$

MNOŽENJE: $z_1 \cdot z_2 = r_1 r_2 \cdot e^{i(\varphi_1 + \varphi_2)}$

DJELJENJE: $\frac{z_1}{z_2} = \frac{r_1}{r_2} \cdot e^{i(\varphi_1 - \varphi_2)}$

POTENCIRANJE: $z = r \cdot e^{i \cdot \varphi} \Rightarrow z^n = r^n \cdot e^{i \cdot \varphi n}$

KORIJEŃOVANJE: $\left(\sqrt[m]{z}\right)_k = \sqrt[m]{r} \cdot e^{i \cdot \frac{\varphi + 2k\pi}{m}}$

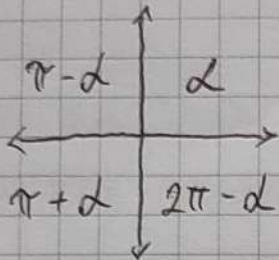
$$r = |z| = \sqrt{x^2 + y^2}$$

$$\cos(\varphi) = \frac{x}{r} \Rightarrow x = r \cos(\varphi)$$

$$\tan(\varphi) = \frac{y}{x} \Rightarrow \varphi \dots$$

$$\sin(\varphi) = \frac{y}{r} \Rightarrow y = r \sin(\varphi)$$

Obratovanje Gl. Argumenta ($\arg(z) = \varphi$)



α	$\tan(\alpha)$
0	0
$\frac{\pi}{6}$	$\frac{\sqrt{3}}{3}$
$\frac{\pi}{4}$	1
$\frac{\pi}{3}$	$\sqrt{3}$
$\frac{\pi}{2}$	$+\infty$

(x, y) - KARTEZNEVE KOORDINATE



(r, φ) - POLARNE KOORDINATE

$$\tan(\alpha) = \frac{y}{x} = \dots = \text{some } \alpha$$

$$\varphi = \{ \text{KVADRANT} \} = \text{kv.} \pm \alpha = \varphi$$

RAČUNSKE OPERACIJE U TRIGONO. OBLIKU:

ZBRAJANJE I ODUZIMANJE NIJE POGODNO!

MNOŽENJE:

$$Z_1 = r_1 (\cos(\varphi_1) + i \sin(\varphi_1))$$

$$Z_2 = r_2 (\cos(\varphi_2) + i \sin(\varphi_2))$$

$$[Z_1 \cdot Z_2 = r_1 r_2 (\cos(\varphi_1 + \varphi_2) + i \sin(\varphi_1 + \varphi_2))]$$

DIJELJENJE:

$$\left[\frac{Z_1}{Z_2} = \frac{r_1}{r_2} (\cos(\varphi_1 - \varphi_2) + i \sin(\varphi_1 - \varphi_2)) \right]$$

POTENCIJANJE:

$$Z^2 = Z \cdot Z = \dots$$

$$[Z^n = r^n (\cos(n \cdot \varphi) + i \sin(n \cdot \varphi))]$$

"DEMOIVRA
FORMULA"

CONTINUED

POTENCNE IMAGINARNE JEDINICE:

$$i^0 = 1$$

$$i^1 = i$$

$$i^2 = -1$$

$$i^3 = i^2 \cdot i = -1 \cdot i = -i$$

$$i^4 = i^2 \cdot i^2 = -1 \cdot (-1) = 1$$

⋮

$$i^{4n} = 1$$

$$i^{4n+1} = i$$

$$i^{4n+2} = -1$$

$$i^{4n+3} = -i$$

Primer:

$$i^{23} = ?$$

$$i^{4 \cdot 5 + 3} = -i$$

SVOJSTVA MODULA KOMPLEKSNIH BROJEVA:

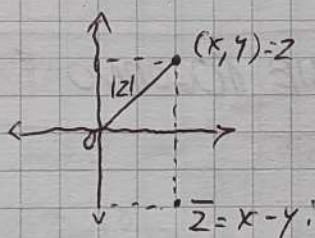
$$|z_1 \cdot z_2| = |z_1| \cdot |z_2|$$

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}$$

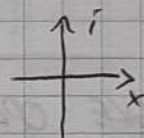
$$|z^n| = |z|^n$$

KOMPLEKSNA (GAUSSOVA) ~~RAVNI~~ RAVNINA:

$$z = x + yi \rightarrow (x, y) \text{ i } x, y \in \mathbb{R}$$



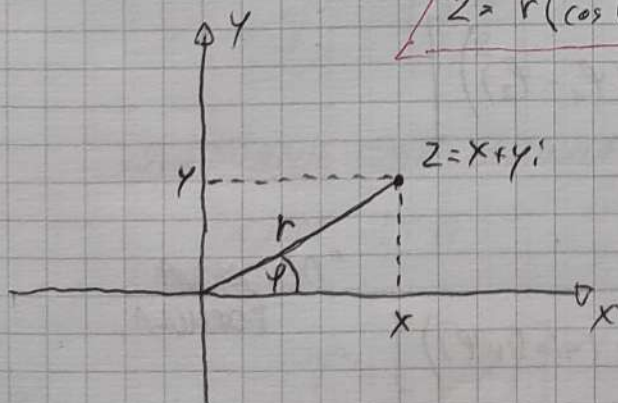
"imaginarna Os."



TRIGONOMETRISKI ZAPIS:

$$z = x + yi \rightarrow z = r \cdot \cos(\varphi) + r \cdot \sin(\varphi) \cdot i$$

$$z = r(\cos(\varphi) + \sin(\varphi)i)$$



r - modul kompleksnog broja

φ - GLAVNI ARGUMENT

$$\varphi \in [0, 2\pi)$$

--- KOMPLEKSNi BROJEVI ---

$$x^2 + 1 = 0$$

$$x^2 = -1 \quad | \sqrt{}$$

$$x_{1,2} = \pm \sqrt{-1} = \pm i$$

i = IMAGINARNA JEDINICA

$\mathbb{C}$ = SKUP KOMPL. BROJEVA = $\{x + y \cdot i \mid x, y \in \mathbb{R}\}$

$$z = x + yi$$

$$x = \operatorname{Re}(z)$$

$$y = \operatorname{Im}(z)$$

$$\operatorname{Re}(z) = 0 = \text{čisto imaginarni}$$

Algebarski Zgibi

Realni Dio

Imaginarni Dio

$\operatorname{Im}(z) = 0$ = čisto realni broj

Kompl. Broj

-- OPERACIJE (ALGEBARSKI IZRAZ) --

1. JEDNAKOST: $z_1 = z_2 \quad \begin{cases} \operatorname{Re}(z_1) = \operatorname{Re}(z_2) \\ \operatorname{Im}(z_1) = \operatorname{Im}(z_2) \end{cases}$

2. MODUL KOMPL. BR.: $|z| = \sqrt{(\operatorname{Re}(z))^2 + (\operatorname{Im}(z))^2}$

3. KOMPLEKSNOM KONJUGIRANI BR.: $z = x + yi \quad \bar{z} = x - yi$

Primer: $z_1 = 2 - 3i \quad z_2 = 4 + i$

$$i^2 = -1$$

4. ~~Zbrajanje~~ ZBRAJANJE: $z_1 + z_2$

$$= 2 - 3i + 4 + i = 6 - 2i$$

5. ODUZIMANJE: $z_1 - z_2$

$$= 2 - 3i - (4 + i) = 2 - 3i - 4 - i = -2 - 4i$$

6. MNOŽENJE: $z_1 z_2$

$$= (2 - 3i)(4 + i) = 8 + 2i - 12i - 3i^2 = 11 - 10i$$

$-3(-1) = 3$

7. KVADRIRANJE: z_1^2

$$= (2 - 3i)^2 = \text{KVADRAT BINOMA} = 2^2 - 2 \cdot 2 \cdot 3i + (3i)^2$$

$$= 4 - 12i + 9i^2 = -5 - 12i$$

$+9(-1) = -9$

8. DIJELJENJE: $\frac{z_1}{z_2}$

$$= \frac{z_1}{z_2} \cdot \frac{\bar{z}_2}{\bar{z}_2} = \frac{2 - 3i}{4 + i} \cdot \frac{4 - i}{4 - i} = \frac{8 - 2i - 12i + 3i^2}{4^2 - i^2} = \frac{5 - 14i}{17}$$

$4^2 - i^2 \rightarrow -1(-1) = 1$

RAZLIKA KVADRATA: $(a - b)(a + b) = a^2 - b^2$

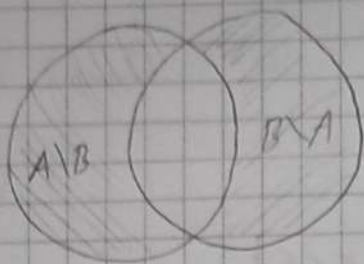
$$= \frac{5}{17} - \frac{14i}{17}$$

KRAJNJI REZULTAT MORA
BITI VIDLJIV $\operatorname{Re}()$ I $\operatorname{Im}()$ dio

$$x + yi$$

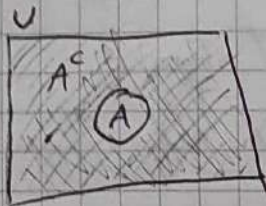
SIMETRIČNA RAZLIKA: $A \Delta B = (A \setminus B) \cup (B \setminus A)$

- KOMUTATIVNA operacija



UNIVERZALNI SKUP: U - univ. s. ; $A \subseteq U$

KOMPLEMENT SKUPA: $A^c = U \setminus A = \{x \in U : x \notin A\}$; $\bar{A}$



KARTEZIJEV PRODUKT: $\{ \}$ - Nizun proizvod elemenata

$$A = \{a, b, c\}$$

$$B = \{1, 2\}$$

() - Uvjeti / Vazun proizvod elemenata

$$A \times B = \{(a, b) \mid a \in A : b \in B\}$$

$$A \times B \neq B \times A$$

$$B \times A = \{(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)\} ; k(B \times A) = 3 \cdot 2 = 6$$

$$k(A) = n$$

$$k(B) = m$$

$$k(B \times A) = n \cdot m$$

$$9. INVOLUTIVNOST: (A^c)^c = A$$

10. PRAVILA APSORPCIJE:

! TEOREMI: Zadano: U ; A, B, C .

$$A \cap (A \cup B) = A ; \\ A \cup (A \cap B) = A$$

1. IDEMPOTENTNOST: $A \cap A = A$; $A \cup A = A$

2. KOMUTATIVNOST $\cap$; $\cup$: $A \cap B = B \cap A$; $A \cup B = B \cup A$

3. ASOCIJATIVNOST $\cap$; $\cup$: $(A \cap B) \cap C = A \cap (B \cap C)$; $(A \cup B) \cup C = A \cup (B \cup C)$

4. DISTRIBUTIVNOST prema $\cup$; obrnuto:

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C) ; A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$

5. DE-MORGANOVE FORMULE: $(A \cap B)^c = A^c \cup B^c$; $(A \cup B)^c = A^c \cap B^c$

6. ...

$$A \cap U = A ; A \cup \emptyset = A$$

7. ...

$$A \cap \emptyset = \emptyset ; A \cup U = U$$

8. SVOJSTVA A^c : $A \cap A^c = \emptyset$; $A \cup A^c = U$

Cent Sheet:

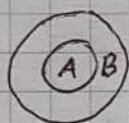
--- Algebra skupova ---

$$A = \{1, 2, 3, 4, 5\} \text{ ili } A = \{x : x \in \mathbb{N} : x < 6\}$$

A - Skup a - Element $\hat{=}$ $a \in A \rightarrow$ element je Skup $\hat{=}$ $\notin \rightarrow$ nije dio Skupa

$B = \{\}$ $\hat{=}$ $\emptyset$ - oznaka za prazan skup

PODSKUP: Ako $\forall$ od A je jednak i element skupa B



$$A \subseteq B \hat{=} \emptyset \subseteq A \hat{=} \emptyset \subseteq X \text{ svaki Skup}$$

PARTITIVNI SKUP: $P(A) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$

$$A = \{1, 2\}$$

KARDINALNI BROJ: $k(\text{Skup}) \hat{=}$ $k(A) = 2 \hat{=}$ $k(P(A)) = 4 = 2^n$

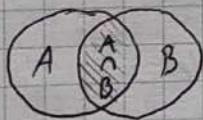
$\hookrightarrow$ Broj elemenata = n

DISJUNKTIVNI SKUPovi: Nema Zajedničkog Elementa $\hat{=}$ $A = \{1, 2\}, B = \{3, 4\}$

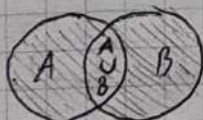
JEDNAKOST SKUPOVA: $A = \{1, a, x\}, B = \{x, 1, a\}, A = B$

!!! OPERACIJE SKUPOVIMA !!!

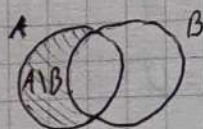
PRESJEK: $A \cap B = \{x : x \in A : x \in B\}$



UNIJA: $A \cup B = \{x : x \in A \text{ ili } x \in B\}$



RAZLIKA: $A \setminus B = \{x : x \in A : x \notin B\}$



- Nije KOMUTATIVNA operacija

OPER. ZAPIS OP. SKUPOVA

AND $\hat{=}$ + $\hat{=}$ $\cap$

OR $\hat{=}$ | $\hat{=}$ $\cup$

NOT $\hat{=}$ - $\hat{=}$ $\setminus$